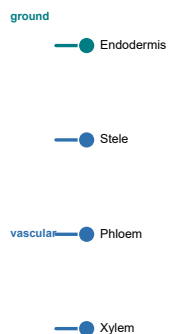
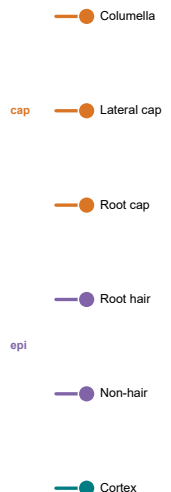
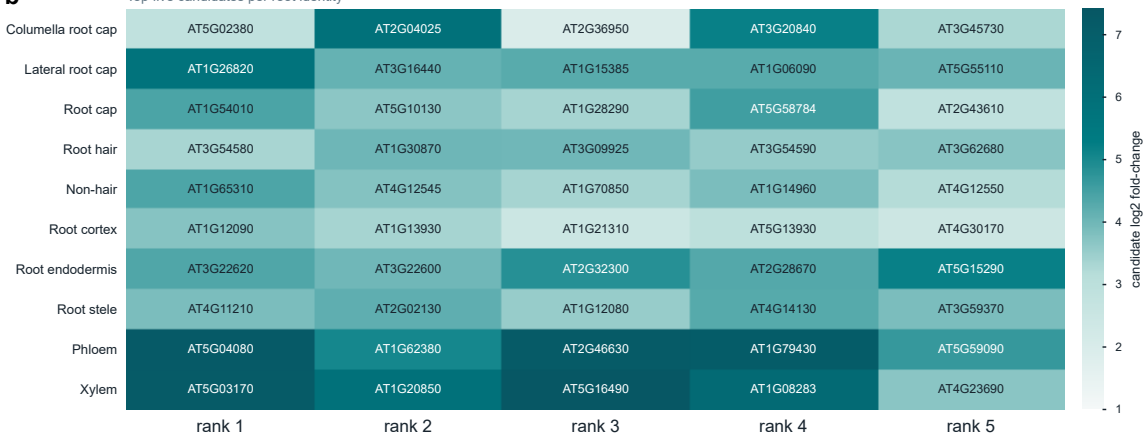


## a Root identity taxonomy

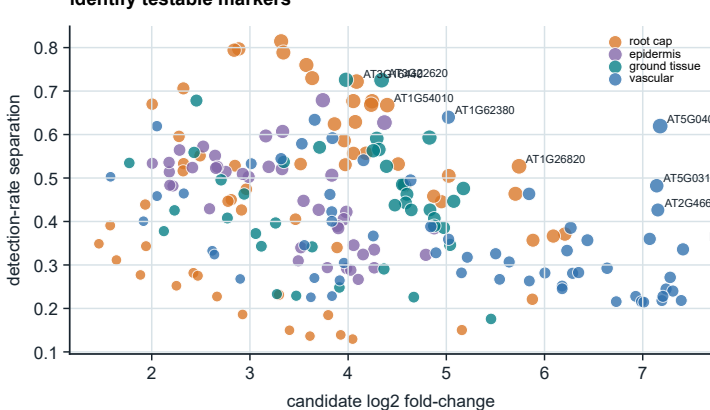


## b State-resolved marker-candidate programs

Top five candidates per root identity



## c Candidate effect size and detection separation identify testable markers



## d Program strength differs across root identities

